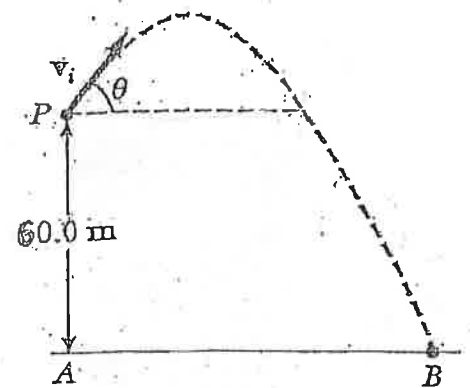


3. A simple, 3.00m long pendulum is released from rest when the support string is at an angle  $28.0^\circ$  to the vertical. What is the velocity of the suspended mass at the bottom of the swing?

4. A particle of mass  $1.50\text{kg}$  is shot from point P, which is located  $60.0\text{m}$  above the ground, as shown below. The particle has an initial velocity  $V_i$  that has a horizontal component of  $28.0\text{m/s}$ . The particle rises to a maximum height of  $15.0\text{m}$  above P.

- a.) What is the vertical component of  $V_i$  ?  
b.) How much work is done by the  $F_g$  force on the particle during its motion from P to B ?  
c.) What is the total final velocity  $V_f$  at point B ?



5. A daring student jumps from a (stationary) hot air balloon with a bungee cord attached to her ankles. The unstretched length of the bungee cord is  $25.0\text{m}$ , the student weighs  $650\text{N}$  and the balloon is  $46.0\text{m}$  above the ground. Assuming that Hooke's Law applies to the bungee cord, calculate the required elastic constant,  $k$ , if the student's lowest point is  $12.00\text{m}$  above the ground.

6. A block of mass  $4.00\text{kg}$  sliding on a horizontal, frictionless surface is attached to one end of a horizontal spring ( $k=100.0\text{N/m}$ ) which has its other end fixed. If the maximum distance the block slides from equilibrium is  $22.0\text{cm}$ , what is the velocity of the block when it is a distance of  $18.0\text{cm}$  from equilibrium? What is the block's acceleration at this point?